Kai Sun

✉ kai.sun1@monash.edu | Google Scholar | LinkedIn

SUMMARY

I am a PhD candidate in Data Science and AI at Monash University under the supervision of Prof. Levin Kuhlmann and Ben Duan. My earlier research focused on algorithmic methods for transportation systems. My primary research focus is now **energy-efficient brain-inspired learning**, with a focus on improving accuracy in image classification through **knowledge distillation** and **neuron design**. I am also exploring how brain-inspired methods can improve the efficiency of large language models. In addition, I have served in student leadership roles for seven years across my undergraduate and master's studies.

EDUCATION

Monash University, Melbourne, Australia
Ph.D. in Information Technology, Data Science and Artificial Intelligence **Expected 2027**
Research: Brain-inspired Neural Network, LLM Reasoning

Hohai University, Nanjing, China
Master of Engineering in Transportation Planning and Management (HD) **2023**
Honors: National Scholarship (0.2%), First Price Scholarship (3 years), Outstanding Graduate/ Student Leader

Guilin University of Electronic Technology, Guilin, China
Bachelor of Engineering in Traffic Engineering (HD) **2020**
Honors: First Price Scholarship, Outstanding Student Leader, Provincial First Prize in a Sci & Tech Competition

SELECTED PUBLICATIONS *Corresponding (or co-corresponding) author

[IJCAI'26] Kai Sun, Peibo Duan*, Yongsheng Huang, Guowei Zhang, Benjamin Smith, Nanxu Gong, Levin Kuhlmann*. **Not All Timesteps Matter Equally: Selective Alignment Knowledge Distillation for Spiking Neural Networks.** *(IJCAI'26) (CORE A*)*.

[ICLR'26] Kai Sun, Peibo Duan*, Yongsheng Huang, Nanxu Gong, Levin Kuhlmann*. **Many Eyes, One Mind: Temporal Multi-Perspective and Progressive Distillation for Spiking Neural Networks.** *International Conference on Learning Representations (ICLR'26) (CORE A*)*.

[ECCV'26] Guowei Zhang, Yun Zhao, Kai Sun, Moein Khajehnejad, Adeel Razi, Dinh Phung, Levin Kuhlmann. **Hi-DREAM: Brain-Inspired Hierarchical Diffusion for fMRI-to-Image Reconstruction via ROI Encoder and VisuAl Mapping.** *European Conference on Computer Vision (ECCV'26) (CORE A*)*.

[IJCAI'25] Kai Sun, Peibo Duan*, Levin Kuhlmann*, Beilun Wang, Bin Zhang. **ILIF: Temporal Inhibitory Leaky Integrate-and-Fire Neuron for Overactivation in Spiking Neural Networks.** *International Joint Conference on Artificial Intelligence (IJCAI'25) (CORE A*)*.

[ECAI'25] Yongsheng Huang, Peibo Duan*, Zhipeng Liu, Kai Sun, Changsheng Zhang, Bin Zhang, Mingkun Xu. **CogniSNN: A First Exploration to Random Graph Architecture Based Spiking Neural Networks with Enhanced Expandability and Neuroplasticity.** *European Conference on Artificial Intelligence (ECAI), 2025 (CORE A)*.

PREPRINTS

[Preprint] Kai Sun, Peibo Duan*, Zhanglu Yan, Guowei Zhang, Yongsheng Huang, Levin Kuhlmann*. **FRAPS: Firing-Rate Alignment for Post-Training Quantization and Structured Pruning of Spiking Neural Networks.** *Preprint*.

[Preprint] Kai Sun, Peibo Duan*, Guowei Zhang, Yongsheng Huang, Nanxu Gong, Levin Kuhlmann*. **Does Spiking Attention Really Attend? Rethinking Self-Attention in Spiking Neural Networks.** *Preprint*.

EXTRACURRICULAR EXPERIENCE

Teaching Associate - Monash University Jan. 2024 - Present
Taught tutorials and assessed coursework for FIT5215 Deep Learning, FIT5225 Cloud Computing and Security, and FIT5197 Statistical Data Modelling.

President of Graduate Student Union - Hohai University May. 2022 - May. 2023
Led graduate-student representatives and organized academic, cultural, and career events.